

Question 1 (7.5 points)

The economy of country MACRO is described by an extend IS-LM model and it is in a short run equilibrium. Suppose that uncertainty due to the Middle East war increases the risk aversion among investors.

- a) Describe the effects of this change on the equilibrium level of output and interest rates. How does the composition of the demand change in the new equilibrium? What about national saving and the real money demand? Explain economically and represent graphically.
- b) Suppose that the central bank wants to return income to the initial level. Which policy intervention should the central bank implement? Compare the composition of the aggregate demand after the central bank intervention with the one prevailing before the change in risk aversion.

Question 2 (7.5 points)

Two BIEM students are discussing about the effects of a tax on bank deposits. Student 1 says *"A tax on bank deposits certainly reduces the money supply but it does not necessarily influence the interest rate: it depends on the behavior of the central bank"*. Student 2 replies: *"No, I don't agree. A tax on deposits reduces the demand for money and, consequently, the interest rate"*. Which of the two students is right? Justify your answer based on the relevant economic theory.

Question 3 (7.5 points)

In the labor market of country Gamma the unemployment rate is at its natural level. Suppose the government implements a reform that increases workers protection by increasing firms firing costs and by providing greater income support for workers who lose their jobs.

- a) After having defined the natural unemployment rate, using the WS-PS model, explain how the natural unemployment rate and the real wage vary in the medium term. Provide a graphical representation and the economic intuition.
- b) Suppose the government decides to combine the previous reform with a strong investment in education and training for workers, which increases their productivity by 25%. How does the labor market equilibrium change in the medium run? What about the potential output? Explain economically.

Question 4 (7.5 points)

The macroeconomic system of country ABC is described by a standard IS-LM-PC model. Assume that expectations are well anchored to the central bank target and that the country is in its medium run equilibrium, denoted by N .

- a) Suppose that the government decides to pass a labor-market reform, which reduces labor protection. What happens in the short and in the medium run if the Central Bank wants to keep inflation at the announced target?
- b) Suppose now that, after the labor reform described in part (a), the government decides to increase government spending G . Can a higher G help the economy reach its new medium-run equilibrium?